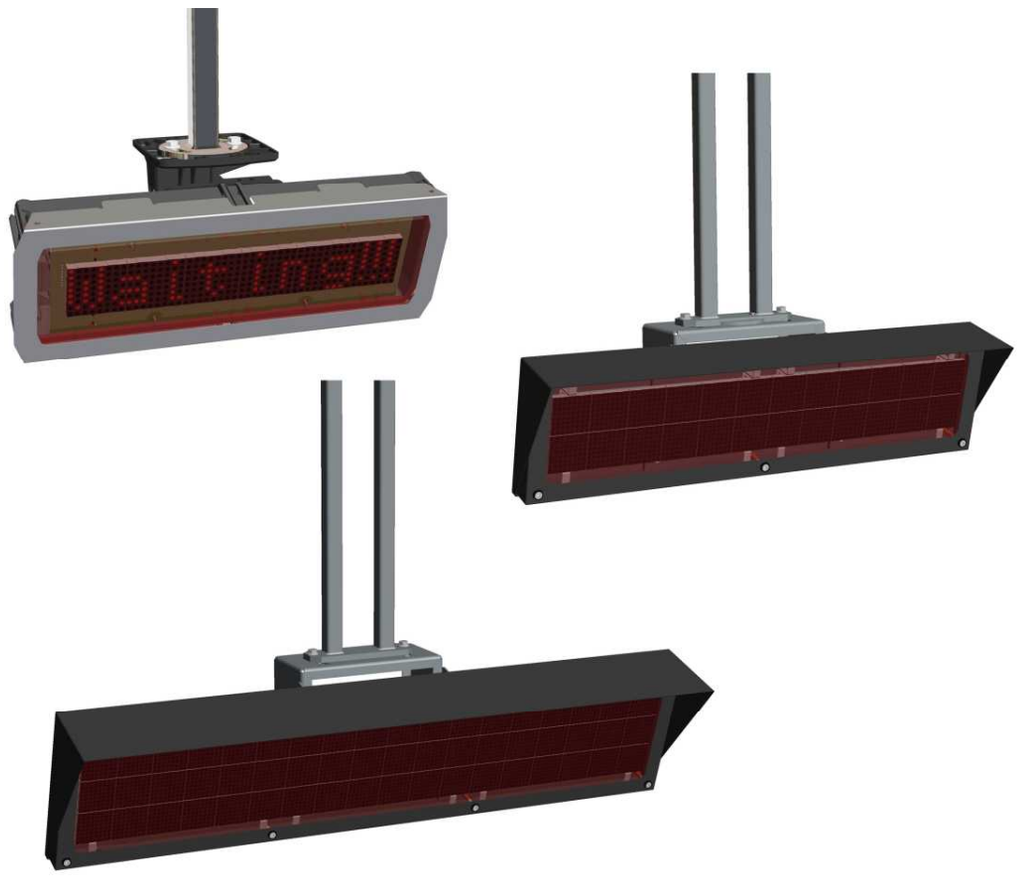
PRODUCT MANUAL

ABUS Load and status display for the crane and trolley

LED matrix display



SERIAL NUMBER:

121-009151

AT A GLANCE:

Checking the requirements: Page 9

Installing the LED matrix display: Page 10

Aligning the LED matrix display: Page 14

Switching on the numeric display for the trolley: Page 17

Overview of tightening torques: Page 18

AN 120204EN002
2024-02-02

Original Operating Instructions

LED MATRIX DISPLAY: VARIOUS SIZES AND VERSIONS

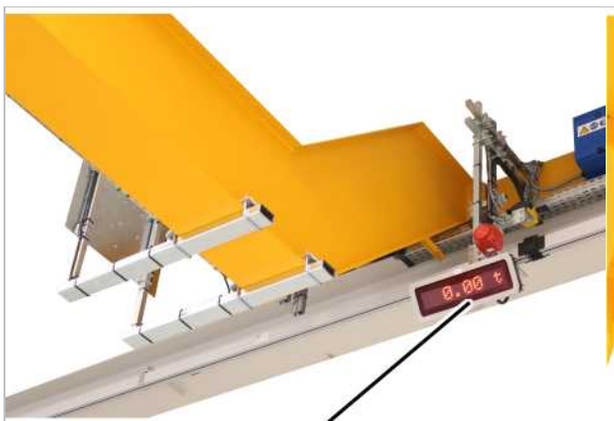
This product manual is applicable for LED matrix displays in various sizes and versions. The work steps described and the technical data will vary according to the size and version of the LED matrix display. The sections of this product manual which do not apply to all LED matrix displays, but are applicable only under certain conditions, are enclosed in a dashed box. At the start of the box, the size and version for which the section is applicable are specified.

If a work step is described in a dashed box:

- At the start of the dashed box text, read the size or version to which this box applies.
- Note page and turn to this first page.
- Based on the images, determine which size or version applies to the crane or trolley at hand.
- Turn back to the page with the associated dashed box for the next work steps.

- The size or version that applies to the crane or trolley in question can also be determined by consulting the scope of delivery or the planning documents.

ATTACHMENT (VERSION)



LED matrix display on the crane



LED matrix display on the wire rope hoist

SIZE



LED matrix display 60 with bracket suspension



LED matrix display 120 with bracket suspension



LED matrix display 180 with bracket suspension

CONTROL (VERSION)



Load indicator system LIS-SV



ABUControl

CONTENTS

GENERAL INFORMATION 5

At the start 5

The LED matrix display 6

Disposing of the LED matrix display .. 8

INSTALLING AND CONNECTING 9

Checking the requirements 9

Installing the LED matrix display 10

Connecting the LED matrix display .. 13

Aligning the LED matrix display 14

Opening and closing the housing of the LED matrix display 15

Switching on the numeric display for the trolley 17

Overview of tightening torques 18

GENERAL INFORMATION

THIS SECTION APPLIES TO ANYONE WHO LIFTS LOADS USING THE CRANE, PERFORMS WORK ON THE CRANE, OR WORKS NEARBY.

AT THE START

USING THIS PRODUCT MANUAL

The following symbols are used in this product manual:

DANGER TO PERSONS!
This warning notice indicates hazards for persons.

DANGER FROM ELECTRIC SHOCK!
This warning notice indicates hazards for persons due to incorrect handling of the electrical system and electricity.

DANGER FROM FALLING SUSPENDED LOAD!
This warning notice describes hazardous situations that can lead to the suspended load falling.

NOTICE OF POSSIBLE DAMAGE!
This notice describes situations in which a component could be damaged.

➔ This is an instruction regarding an action and prompts you to perform a work step.

- This is the result of an action and describes what happens on the device.

– This is a list.

ONLY WITH...

A section that appears in a box framed by a dashed line is only applicable for certain types, versions or options. The condition to which the section applies is specified at the beginning in the heading “Only with...”.

INSTRUCTIONS FOR THE PRODUCT MANUAL

Read the product manual through carefully before beginning work. Also be sure to observe additional product manuals for accessories and components.

Afterwards, keep the product manual available in the vicinity of the crane. It must be accessible to all who work with or on the crane.

In the case of reselling or hiring out, always provide the product manual together with the crane.

PROPER USE

The LED matrix display is used to display the suspended load measured by a load measuring system (load indicator system LIS-SV or ABUControl).

In addition to the suspended load, messages from the load indicator system LIS-SV or ABUControl are displayed.

– The displayed load is not a calibrated measurement and is only an approximate estimate of the suspended load.

– In the load indicator system LIS-SV and ABUControl: The suspended load is continuously measured and displayed.

REGULATIONS

The installation has been built and tested according to European standards, rules and regulations applicable at the time of its manufacture. The conventions applied during design and construction are specified in the Declaration of Conformity or the Declaration of Incorporation. The conventions must also be observed for installation, operation, inspection and maintenance, as must the applicable occupational health and safety regulations.

DANGER TO PERSONS!
Non-observance of the regulations can result in the death of persons or serious accidents.

To ensure that the work can be performed safely, thorough instruction in regard to this product manual and the regulations is necessary.

Which provision applies in individual cases will greatly depend on the crane's use and the national regulations. Review and observe all applicable and current regulations, including the occupational health and safety regulations. Refer also to the Declaration of Conformity or Declaration of Incorporation.

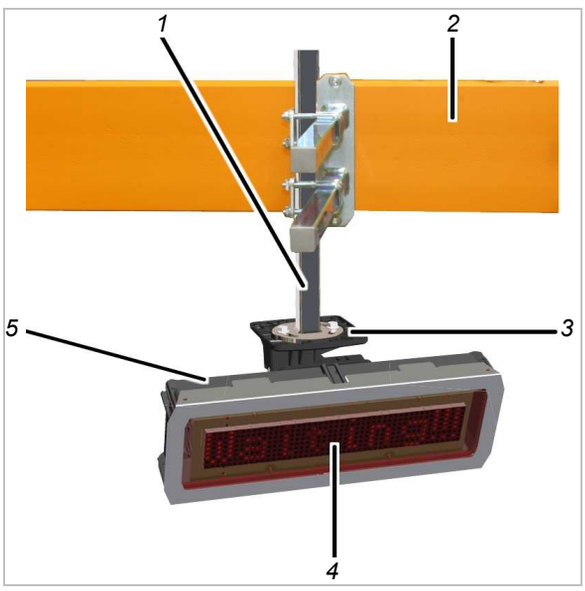
WARRANTY

- ABUS assumes no liability for damages that were caused by improper use, inadequately trained personnel, incorrect performance of work, nor for any changes, reconstruction or other modification of the crane or crane components which were not approved by ABUS.
- Any warranty claim becomes invalid if components were modified without consent, the crane or crane components were installed, used or serviced in any way other than that described in this product manual, or if parts other than original ABUS replacement parts were used.
- The safe operation of the crane or crane components is only guaranteed if original ABUS replacement parts are used.

THE LED MATRIX DISPLAY

DESCRIPTION OF THE DEVICE

The figure shows the LED matrix display 60. The LED matrix displays 120 and 180 are not significantly different.



- 1: Mounting tube
- 2: Crane or trolley end carriage
- 3: Bracket
- 4: LED matrix display
- 5: Housing

PERFORMANCE FEATURES

The LED matrix display:

- In normal operation, the LED matrix display on the end carriage of the crane shows the load currently suspended on the load hook.
- In addition, messages from the load indicator system LIS-SV or the crane control ABUControl are displayed.
- The LED matrix display can be rotated and tilted so that the crane operator can easily see the displayed information.
- The LED matrix display can be connected to crane installations with the load indicator system LIS-SV via a Q signal interface, or to crane installations with the crane control ABUControl via CAN bus, or via Modbus with a wind measuring system.
- The LED matrix display can display most European languages through its extended character set.
- In the load indicator system LIS-SV: The LED matrix display can when fitted to a trolley also show the numbers “1” and “2” to indicate the assignment of the trolley.
- In ABUControl: The LED matrix display can display all status and error messages of the crane control as scrolling text.
- With the wind measuring system: The LED matrix display can show the wind speed in various units.

TECHNICAL DATA

LED matrix display:

	60	120	180
LED matrix	8x48 dots	16x96 dots	24x144 dots
Character size	60 mm	120 mm	180 mm
Reading distance	20 to 30 m	50 to 60 m	70 to 80 m
Maximum rotation	±45°	±28°	±28°
Maximum inclination	45°	45°	45°

Dimensions and weights:

	60	120	180
Width x height	467 x 166 mm	801 x 218 mm	1167 x 279 mm
Depth	74 mm	125 mm	156 mm
Weight	1.9 kg	7.1 kg	12.7 kg

Ambient conditions for operation:

Ambient temperature (for normal operation)	-5 °C to +55 °C
Housing	IP 54

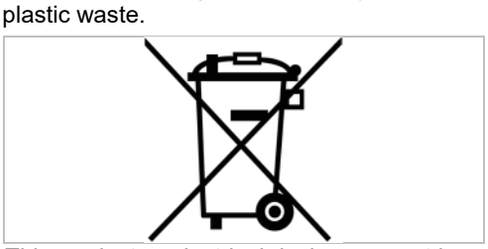
Electrical connection

	60	120	180
Operating voltage	24 V DC +/-20%		
Current consumption	200 mA	400 mA	900 mA
Energy consumption	4.8 W	9 W	20 W
Q signals connection cable	5G1 mm²		
CAN bus/Modbus connection cable	4 x 2 x 0.1 mm²		

DISPOSING OF THE LED MATRIX DISPLAY

When disposing of the LED matrix display:

- ➔ Dismantle the components of the LED matrix display as far as possible.
- ➔ Observe local regulations concerning disposal and recycling.
- ➔ Dispose of the individual parts sorted by material in an environmentally sound manner.
  - – Remove electronic assemblies and dispose of in accordance with local regulations for scrap electronic parts.
  - – The plastic housing should be recycled with plastic waste.



This product or electrical device may not be disposed of at the end of its service life with regular domestic waste.

INSTALLING AND CONNECTING

THIS SECTION APPLIES TO ANYONE WHO WORKS ON THE CRANE PRIOR TO ITS USE

The end user of the crane is responsible for the proper qualifications of the commissioning personnel.

DANGER TO PERSONS!
Persons can be injured if the crane is incorrectly put into operation.

If personnel other than that of the ABUS company are employed to perform the crane commissioning, it is the end user's responsibility to ensure that these persons are adequately qualified. Follow the procedures described here precisely.

Examples of qualified persons:

- Persons with comprehensive knowledge from specialist training in engineering and in the electrical systems of cranes.
- Persons with sufficient experience in the operation, installation and maintenance of cranes.
- Persons with comprehensive knowledge regarding the relevant technical rules, directives and safety regulations applicable in the respective country.
- Persons receiving regular training from ABUS.

ABUS assumes no liability for damage due to incorrectly performed commissioning work done by unqualified personnel.

ABUS recommends having the commissioning work performed by the ABUS assembly team.

CHECKING THE REQUIREMENTS

The following requirements must be met in order for the LED matrix display to be installed:

- The crane should be equipped with a crane control ABUControl or a load indicator system LIS-SV.

INSTALLING THE LED MATRIX DISPLAY

The LED matrix display is installed on the end carriage of the crane or the trolley.

In the case of an LED matrix display on a crane: The LED matrix display is usually installed on the end carriage on which the crane panel is installed or on that which is closest to the crane panel. It can also be installed on the end carriage that is on the opposite side.

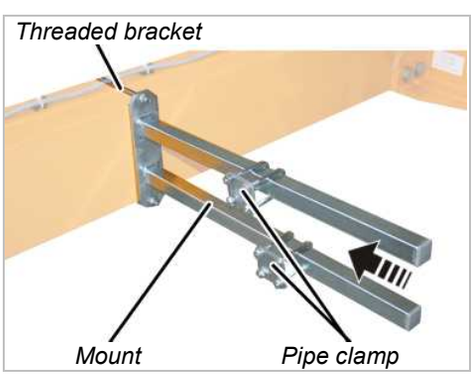
With LED matrix display on the wire rope hoist:
The LED matrix display is mounted on the trolley frame or the end carriage, depending on the design of the wire rope hoist.

In the case of an LED matrix display on a GM modular wire rope hoist: The LED matrix display is installed in the housing cover of the wire rope hoist.

In the following, the assembly is shown on a crane. The installation on a wire rope hoist does not differ significantly.

ONLY FOR THE LED MATRIX DISPLAY 60

INSTALLING THE MOUNT

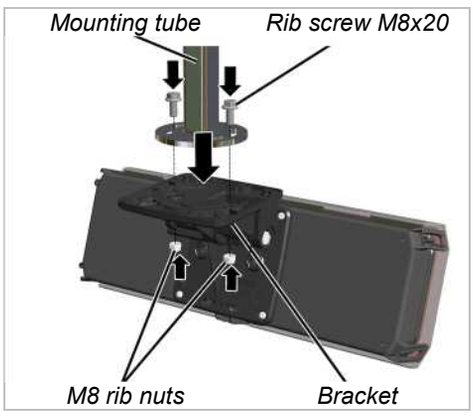


- Attach the mount to the end carriage on which the crane panel is installed.
- Bolt the mount onto the threaded bracket.

Threaded bracket	Tightening torque
M8	25 Nm
M10	50 Nm
M12	75 Nm

- Slide on the two pipe clamps and secure them.

INSTALLING THE MOUNTING TUBE ON THE LED MATRIX DISPLAY



- Bolt the mounting tube with rib screws M8x20 (2x) and self-locking nuts M8 (2x) onto the bracket of the LED matrix display. 5 Nm.

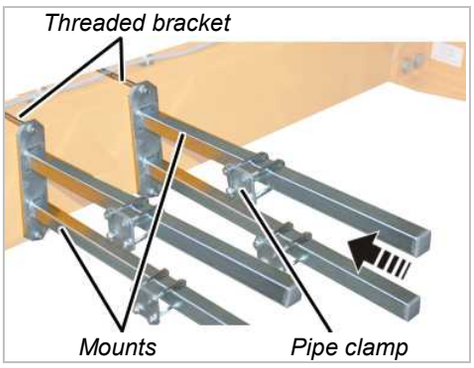
INSTALLING THE MOUNTING TUBE ON THE MOUNT



- Push the mounting tube with LED matrix display into the pipe clamps from below.
- Slide the mounting tube up or down so that the LED matrix display (at the bottom of the mounting tube) can subsequently be read with ease.
- Tighten the pipe clamps. 15 Nm.

ONLY FOR LED MATRIX DISPLAY 120 AND 180

INSTALLING THE MOUNT



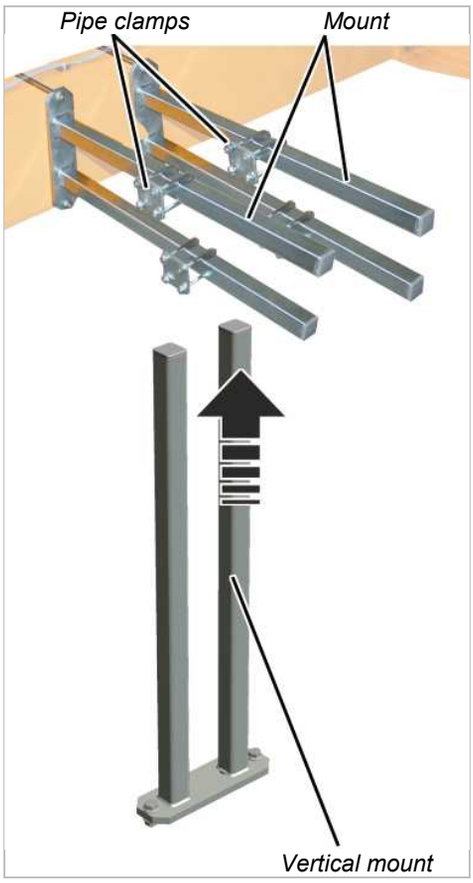
➔ Fasten the two mounts to the end carriage on which the crane panel is installed.

➔ Fasten the mount to the respective threaded bracket.

Threaded bracket	Tightening torque
M8	25 Nm
M10	50 Nm
M12	75 Nm

➔ Slide on the two pipe clamps and secure them.

INSTALLING THE VERTICAL MOUNT



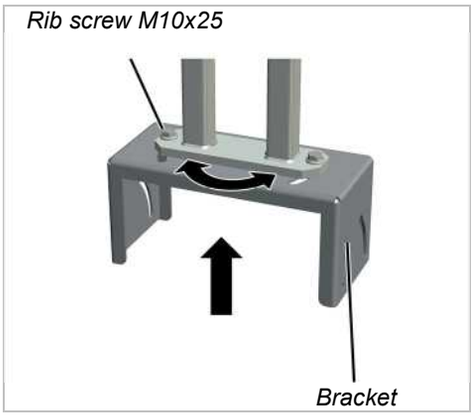
➔ Insert the vertical mount into the four pipe clamps from below.

➔ Slide the vertical mount up or down so that the LED matrix display (at the bottom of the vertical mount) can subsequently be read with ease.

When pushing the mount down, the LED matrix display must not be more than maximally 600 mm below the two horizontal mounts.

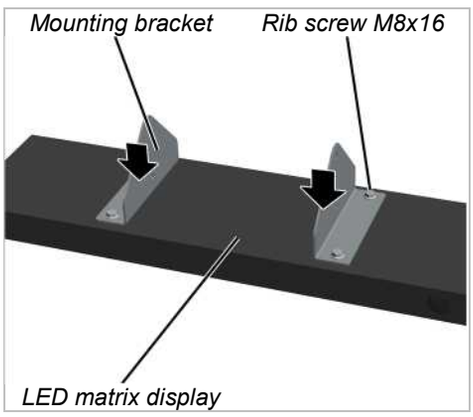
➔ Tighten the pipe clamps. 15 Nm.

INSTALLING THE BRACKET



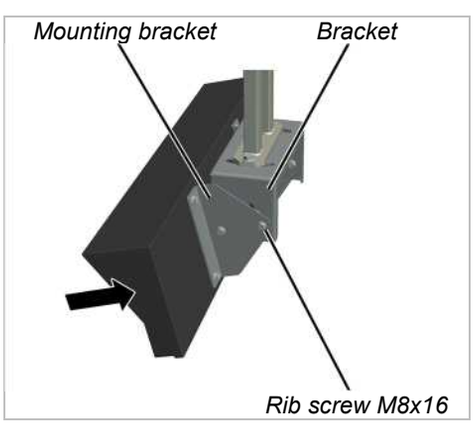
- Hold the bracket under the mount.
- First, loosely screw the bracket to the vertical mount with two M10x25 rib screws and two M10 rib nuts.
- Rotate the bracket so that the LED matrix display can be easily read later.
- Tighten the rib screws. 50 Nm.

BOLTING ON THE LED MATRIX DISPLAY



- Place the two mounting brackets on the back of the LED matrix display.
- Bolt on the mounting brackets (2x) with rib screws M8x16 (4x). 40 Nm.

INSTALLING THE LED MATRIX DISPLAY ON THE BRACKET



- Slide the LED matrix display with the mounting brackets onto the bracket from the inside.
- First, loosely screw on the mounting bracket with the four M8x16 rib screws and four M8 rib nuts.
- Tilt the LED matrix display so that it can easily be seen later.
- Tighten the rib screws. 40 Nm.

CONNECTING THE LED MATRIX DISPLAY

The LED matrix display is connected to a DC power pack in the panel. A separate DC power pack is provided for this purpose or a DC power pack that is also used for other devices.

ONLY FOR THE LOAD INDICATOR SYSTEM LIS-SV

CONNECTING THE LED MATRIX DISPLAY

TAKE NOTE OF THE ASSIGNMENT!
If the assignment is interchanged, the load indicator system LIS-SV can be irreparably damaged.

Take note of and check the assignment!

Take note of the wiring assignment and connect the LED matrix display.

For further details: see the wiring diagram.

Terminal	Conductor
Q signal	3
GND	4
24 V	1
0 V	2

Make sure that the cable is not installed under tension and that it is laid away from the motor cable.

ONLY WITH ABUCONTROL

CONNECTING THE LED MATRIX DISPLAY

Route the RJ-45 connection cable from the control to the LED matrix display in accordance with the wiring diagram and connect.

The RJ-45 connection cable conveys the operating voltage for the LED matrix display and the data signal.

Make sure that the cable is not installed under tension and that it is laid away from the motor cable.

Connect the RJ-45 connection cable to the control on the CAN distributor in accordance with the wiring diagram. For cable lengths 5 m or more, connect on the transfer module.

It doesn't matter which socket on the CAN distributor the connection cable is connected to.

SETTING THE TERMINATING RESISTOR

Depending on the length of the connecting cable and the number of trolleys, it may be necessary to switch on the terminating resistor. See "Setting the terminating resistor in the CAN bus network" in the ABUControl product manual.

SETTING THE DEVICE ADDRESS

Normally the device addresses of the LED matrix displays are already set or are automatically provided.

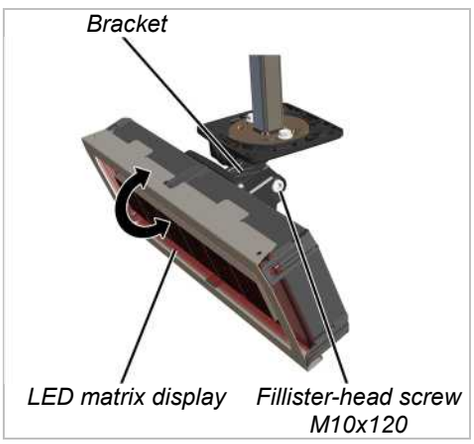
If the device address is to be entered by hand or the automatically provided device address is deleted and is to be reentered: See “Overview of device addresses” in the ABUControl product manual.

ALIGNING THE LED MATRIX DISPLAY

ONLY FOR THE LED MATRIX DISPLAY 60

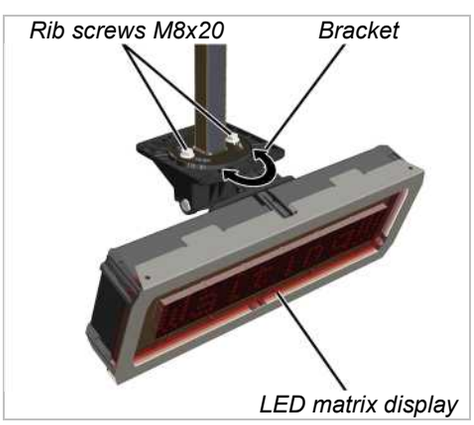
With the LED matrix display 60, the bracket can be tilted in 5° steps and continuously swivelled.

TILTING THE LED MATRIX DISPLAY



- Release the fillister-head screw M10x120.
- Tilt the LED matrix display into the desired position.
- Screw on the fillister-head screws M10x120. 6 Nm.

ROTATING THE LED MATRIX DISPLAY

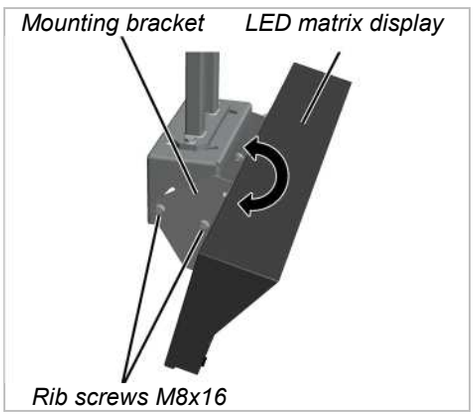


- Release the M8x20 rib screws (2x).
- Rotate the LED matrix display into the desired position.
- Tighten the rib screws. 5 Nm.

ONLY WITH LED MATRIX DISPLAY 120 AND 180

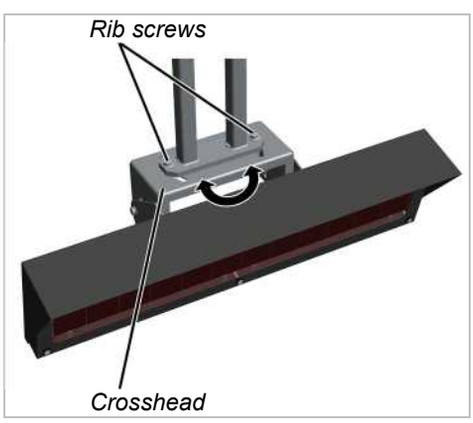
The LED matrix display 120 and 180 can be continuously tilted and swivelled.

TILTING THE LED MATRIX DISPLAY



- Release the M8x16 rib screws (4x).
- Tilt the LED matrix display as required.
- Tighten the rib screws. 40 Nm.

ROTATING THE LED MATRIX DISPLAY



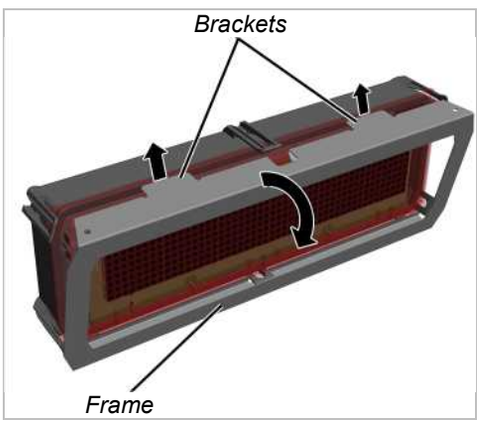
- Release the M10x25 rib screws (2x).
- Rotate the LED matrix display as required.
- Tighten the rib screws. 50 Nm.

OPENING AND CLOSING THE HOUSING OF THE LED MATRIX DISPLAY

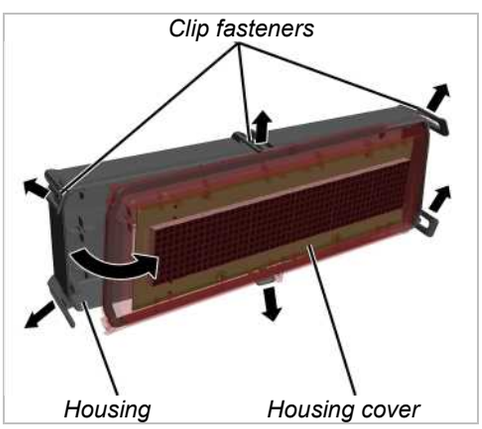
In certain circumstances, the housing of the LED matrix display may need to be opened (for example, to connect the connecting cables or to change the settings).

ONLY FOR THE LED MATRIX DISPLAY 60

OPENING THE HOUSING



➔ Pull the brackets on the frame upward and fold the frame toward the front from the housing cover.

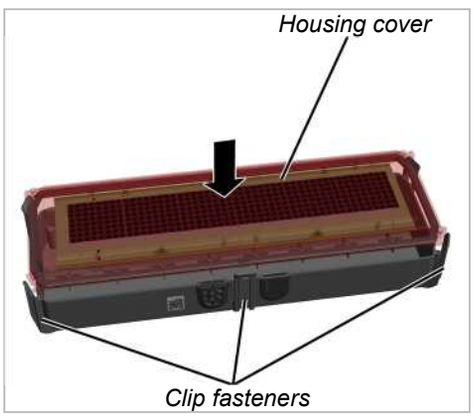


➔ First press the left, then the middle and finally the right clip fastener slightly outward and fold out the housing cover toward the side.

In doing this, make sure the cable is not pinched or pulled.

The circuit board is located in the housing cover, whereas the cable is led out of the bottom part of the housing.

CLOSING THE HOUSING

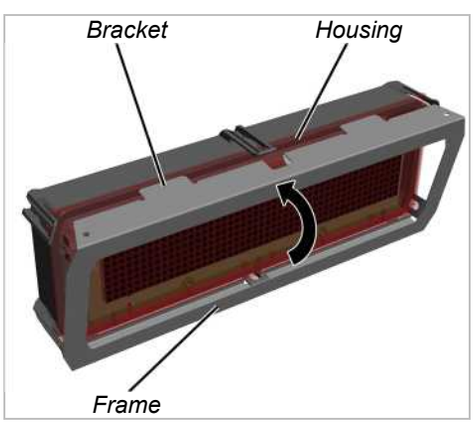


➔ Place all four corners of the housing cover onto the housing.

In doing this, make sure the cable points downward and is led out of the bottom part of the housing.

➔ Latch the housing cover into all six clip fasteners at the same time.

In doing this, make sure the cable is not pinched!



➔ Lay the frame against the lower edge of the housing.

Make sure that the recesses fit and the brackets on the frame point upward.

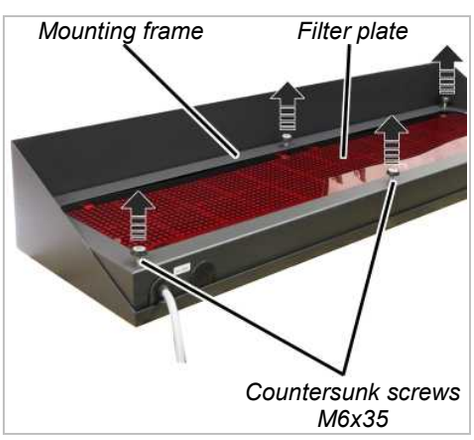
➔ Press the frame onto the housing cover.

ONLY FOR LED MATRIX DISPLAY 120 AND 180

OPENING THE SHEET STEEL CASING

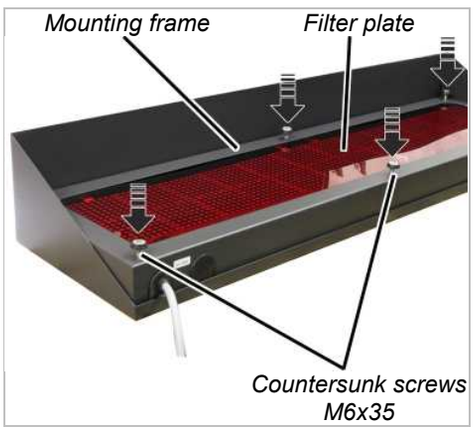
DANGER TO PERSONS DUE TO THE HEAVY WEIGHT OF THE COMPONENT!
The mounting frame is heavy. If it is unscrewed with the LED matrix display fitted, the entire mounting frame with the circuit board and LED matrix modules may fall down and injure persons.

Completely remove the LED matrix display before opening or secure the mounting frame before unscrewing.



- ➔ Completely remove the LED matrix display.
- ➔ Unscrew the six or eight M6x35 countersunk screws.
- ➔ Remove the mounting frame with the filter plate, circuit board and LED matrix modules.

CLOSING THE SHEET STEEL CASING



- ➔ Insert the circuit board with LED matrix modules into the housing.
- ➔ Insert the filter plate and mounting frame.
- ➔ Screw in the M6x35 countersunk screws with thread locking agent (medium strength). 3 Nm.

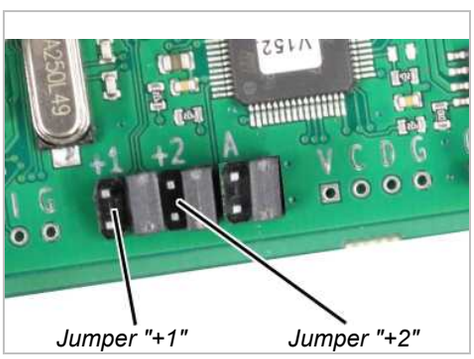
SWITCHING ON THE NUMERIC DISPLAY FOR THE TROLLEY

ONLY FOR THE LOAD INDICATOR SYSTEM LIS-SV

The LED matrix display can when fitted to a trolley also show the numbers "1" and "2" to indicate the assignment of the trolley.

The figure shows the location of the jumper on the LED matrix display 120 and 180. On the LED matrix display 60, the jumper is located elsewhere but is labelled identically.

SWITCHING ON THE NUMERIC DISPLAY



➔ For the display of number "1": plug jumper "+1" into position "+1" (the jumper plugs in directly under the "+1" label).

Jumper "+2" remains to the right of position "+2" and jumper "A" remains to the right of position "A".

➔ For the display of number "2": plug jumper "+2" into position "+2" (the jumper plugs in directly under the "+2" label).

Jumper "+1" remains to the right of position "+1" and jumper "A" remains to the right of position "A".

OVERVIEW OF TIGHTENING TORQUES

THREADED BRACKET



Threaded bracket	Type and size	Number	Tightening torque
M8	Rib nut M8	2x	25 Nm
M10	Rib nut M10	2x	50 Nm
M12	Rib nut M12	2x	75 Nm

PIPE CLAMPS ON THE ROPE GUIDE RUNNER



- Rib nut M8
- 4x per pipe clamp
- 15 Nm

SWIVEL JOINT ON THE MOUNTING TUBE



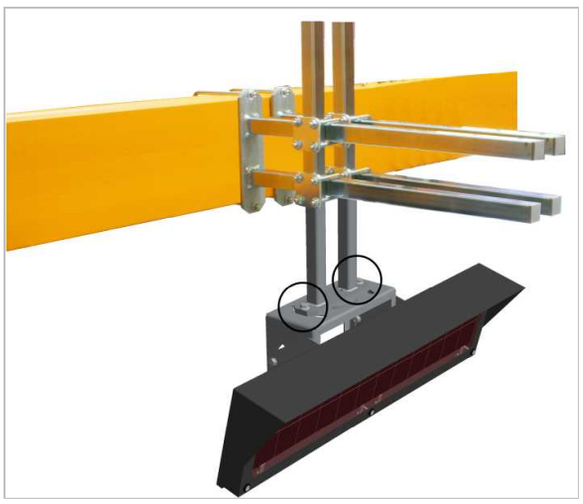
- Rib screw M8x20 with self-locking nut M8
- 2x
- 5 Nm

LED MATRIX DISPLAY ON THE BRACKET



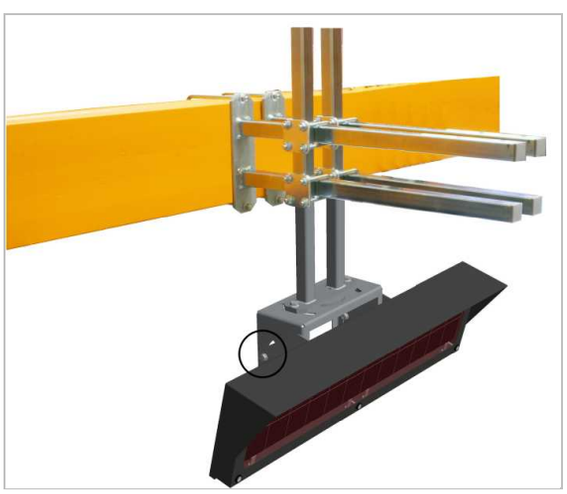
- Fillister-head screw M10x120
- 1x
- 6 Nm

BRACKET ON VERTICAL SQUARE TUBE



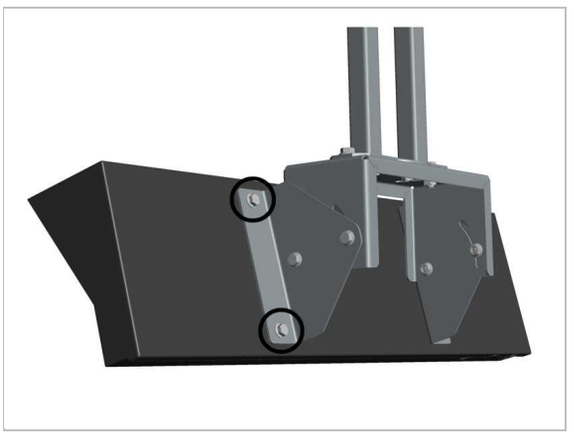
- Rib screw M10x25 with rib nut M10
- 2x
- 50 Nm

LED MATRIX DISPLAY ON THE CROSSHEAD



- Rib screw M8x16 with rib nut M8
- 4x
- 40 Nm

MOUNTING BRACKETS ON THE LED MATRIX DISPLAY



- Rib screw M8x16
- 4x
- 40 Nm

ABUS Kransysteme GmbH
Sonnenweg 1
51647 Gummersbach Germany
Tel. 0049 – 2261 – 37-0
Fax. 0049 – 2261 – 37-247
info@abus-kransysteme.de

Distribution or reproduction of this document, or the utilisation and disclosure of its contents without express consent is prohibited. Violations will result in liability for damages. All rights are reserved in regard to patent issuance or registered designs.

AN 120204EN002
2024-02-02